Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2022-P2-Model)** **Date: 29-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-28** **Max. Marks: 186**

KEY SHEET

MATHEMATICS

1	**2**	2	**8**	3	**2**	4	**7**	5	**6**	6	**6**
7	**2**	8	**4**	9	**AD**	10	**ABCD**	11	**AB**	12	**ABC**
13	**CD**	14	**ABC**	15	**D**	16	**A**	17	**B**	18	**A**

PHYSICS

19	**2**	20	**8**	21	**9**	22	**6**	23	**3**	24	**5**
25	**6**	26	**1**	27	**BCD**	28	**BD**	29	**BCD**	30	**BC**
31	**ABC**	32	**ABCD**	33	**A**	34	**B**	35	**C**	36	**A**

CHEMISTRY

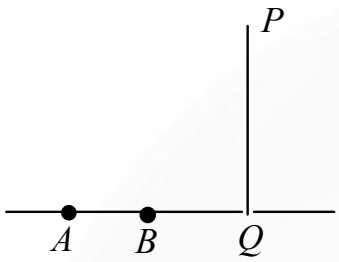
37	**6**	38	**7**	39	**4**	40	**8**	41	**6**	42	**9**
43	**6**	44	**3**	45	**ABCD**	46	**CD**	47	**BD**	48	**ABC**
49	**BC**	50	**AB**	51	**D**	52	**C**	53	**B**	54	**C**

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SOLUTIONS

MATHEMATICS

1. As A, Q, B are collinear $\begin{vmatrix} w & \bar{w} & 1 \\ a & \bar{a} & 1 \\ b & \bar{b} & 1 \end{vmatrix} = 0$



$$\Rightarrow \begin{vmatrix} w & \bar{w} & 1 \\ a & 1/a & 1 \\ b & 1/b & 1 \end{vmatrix} = 0 \qquad \left[\because a\bar{a} = b\bar{b} = 1\right]$$

$$\Rightarrow w\left(\frac{1}{a} - \frac{1}{b}\right) + \bar{w}(b-a) + \left(\frac{a}{b} - \frac{b}{a}\right) = 0$$

$$\Rightarrow \frac{w}{ab} + \bar{w} - \frac{a+b}{ab} = 0 \qquad \Rightarrow \bar{w} = \frac{a+b-w}{ab}$$

Also , as $PQ \perp AB$

$$\frac{a-b}{a-\bar{b}} + \frac{w-z}{\bar{w}-\bar{z}} = 0 \Rightarrow -ab + \frac{w-z}{\bar{w}-\bar{z}} = 0$$

$$\Rightarrow ab\left(\bar{w}-\bar{z}\right) = w - z$$

$$\Rightarrow ab\left(\frac{a+b-w}{ab}\right) - ab\bar{z} = w - z$$

$$\Rightarrow w = \frac{1}{2}(a+b+z-ab\bar{z}) \qquad \Rightarrow \frac{4w}{a+b+z-ab\bar{z}} = 2$$

2. $f(\alpha_i) = \tan\alpha_i \qquad e^{-\alpha_i}\left(f(\alpha_i) + \frac{1}{f(\alpha_i)}\right) = \frac{2e^{-\alpha_i}}{\sin 2\alpha_i} = 2$

$$\sum_{i=1}^{4} e^{-\alpha_i}\left(f(\alpha_i) + \frac{1}{f(\alpha_i)}\right) = 8$$

3. $P_n - P_{n+1} = \frac{1}{I_1.I_2....I_n} - \frac{1}{I_1.I_2..I_nI_{n+1}}$

$$\Rightarrow P_n - P_{n+1} = \frac{(I_{n+1}-1)}{I_1.I_2....I_nI_{n+1}} = \frac{1}{I_{n+1}};$$

$$H_{n+1} - H_n = \left(\frac{1}{I_1} + \frac{1}{I_2} + ... + \frac{1}{I_n} + \frac{1}{I_{n+1}}\right) - \left(\frac{1}{I_1} + \frac{1}{I_2} + ... + \frac{1}{I_n}\right)$$

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$$\Rightarrow H_{n+1}-H_n=\frac{1}{I_{n+1}}\Rightarrow P_{n+1}+H_{n+1}=P_n+H_n$$

$$\Rightarrow P_{n+1}+H_{n+1}=P_1+H_1=2,\forall n\in N$$

4. $$\int_0^{\pi}\frac{2\sin\theta+3\cos\theta-3}{13\cos\theta-5}d\theta=2\int_0^{\pi/2}\frac{2\sin 2x+3\cos 2x-3}{13\cos 2x-5}dx$$

$$=2\int_0^{\pi/2}\frac{\sin x}{2\cos+3\sin x}dx$$

To compute the above integral we want to write sin x as a linear combination of the denominator and its derivative:

$$=2\int_0^{\pi/2}\frac{\sin x}{\cos x+3\sin x}=2\int_0^{\pi/2}\frac{-\frac{1}{13}\left[-3(2\cos x+3\sin x)+2(3\cos x-2\sin x)\right]}{2\cos x+3\sin x}$$

$$=-\frac{2}{13}\left[\int_0^{\pi/2}(-3)+2\int_0^{\pi}\frac{-2\sin x+3\cos x}{2\cos x+3\sin x}\right]\qquad=-\frac{2}{13}\left[-\frac{3\pi}{2}+2\log(3\sin x+2\cos x)_0^{\pi/2}\right]$$

$$=-\frac{2}{13}\left[-\frac{3\pi}{2}+2\log\frac{3}{2}\right]=\frac{3\pi}{13}-\frac{4}{13}\log\frac{3}{2}$$

5. It exists only when p > 3 & q ≥ 3 [p + q] = 6

6. We use LHS to refer to the left hand side of the equation and RHS to refer to the right hand side. Let the degree of f be k, If k ≥ 2. LHS has degree k^2 and RHS has degree 2k, with equality only at k = 2. Thus, the degree is 2.

Let f(x) = ax^2+bx+c. The LHS coefficient of x^4 is a^3, while the RHS coefficient of $x^4 is a^2$; equality gives a = 1. So f(x) = x^2+bx+c. In addition, note that f(x)

Divides $f(x)^2-f(x)=f(f(x))=f(x)^2+bf(x)+c$, forcing c = 0.

Now, the condition given in the problem becomes

$$f(x)^2+bf(x)+f(x)=f(x)^2\rightarrow b=-1$$

Hence, f(x) = x^2-x, and f(3) = 6.

7. For any a, a | 0, so any f(a) | f(0), which gives us f(0) = 0. For any b, 1 | b, f(1) | f(b)| for any b, so f(1) = 1. Similarly, f(2) = 2, f(3) = 3 since we get that f(2) must divide 6 elements in the range and f(3) must divide 4 elements in the range. this then forces f(6) = 6 since 2 and 3 must divide f(6). f(5) must divide f(10), giving the possibilities (4, 8), (5, 10). In the former case this forces f(4) = 5, f(8) = 10, contradicting the given condition in the case (a, b) = (2, 4). In the latter case, f(4) = 4, f(8) = 8 and f(9) = 9 since it is divisible by 3, The only freedom we are afforded is that we can exchange 7 and 11 if we want to, giving 2 distinct functions.

8. Since $f(-\infty)=f(\infty)=+\infty, f(1)<0, f(3)>0$ and f(5) < 0, polynomial f has a real zero in each of the intervals $(-\infty,1),(1,3),(3,5),(5,\infty)$ that is four in total. Δ

We now give the announced proof of the fact that every polynomial has a complex root. This fundamental theorem has many different proofs. The proof we present is, although more difficult than all the previous ones. still next to elementary. All imperfections in the proof are made in purpose.

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9. $\overline{AB}\times\overline{AC}=(\bar{i}+\bar{j}-3\bar{k})\times(3\bar{i}+3\bar{j}-6\bar{k})=3\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\1&1&-3\\1&1&-2\end{vmatrix}=3(\bar{i}-\bar{j})$

Area of $\Delta ABC=\frac{1}{2}\left|\overline{AB}\times\overline{AC}\right|=\frac{1}{2}(3\sqrt{2})=\frac{3}{\sqrt{2}}$ sq. units

$\therefore\frac{1}{2}\left|\overline{BC}\right|$(Altitude through A) $=\frac{3}{\sqrt{2}}$ and $\left|\overline{BC}\right|=\left|2\bar{i}+2\bar{j}-3\bar{k}\right|=\sqrt{17}$

$\Rightarrow$ Altitude through A $=\frac{3\sqrt{2}}{17}$

Also $\left|\bar{a}\bar{b}\bar{c}\right|=\begin{vmatrix}1&-2&3\\2&-1&0\\4&1&-3\end{vmatrix}=9>0$

$\therefore\bar{a}\bar{b}\bar{c}$ are non-coplanar and forms a R.H system.

10.

11. Centers of given circle are $C_1=(1,0),C_2=(-3,0)$ and radii are $r_1=1,r_2=3$ respectively.
They are touching each other externally at origin.
y-axis is a common tangent. x – axis is the line of centers.
External centre of similitude = (3, 0)

y = m (x – 3) is tangent $\Rightarrow d=r\Rightarrow m=\pm\frac{1}{\sqrt{3}}$

$\Rightarrow\Delta ABC$ is isosceles with vertical angle 60° and hence ΔABC is equilateral.
$\therefore G=S=1=(1,0)$

12. Given, $z_1=a+ib,z_2=c+id,\omega_1=a+ic,\omega_2=b+id$

Given, $|z_1|=|z_2|=1\therefore a^2+b^2=1$ ….(i) and $c^2+d^2=1$

$z_1\bar{z}_2=(a+ib)(c-id)=ac+bd+I(bc-ad)$

Again, Re $(z_1\bar{z}_2)=0\Rightarrow ac+bd=0\Rightarrow ac=-bd$

$\therefore\frac{a}{d}=\frac{b}{-c}=k$ (say) ….(iii) $\therefore$ a = kd and b = -kc

From (i), $a^2+b^2=1\therefore k^2d^2+k^2c^2=1$ or $k^2(c^2+d^2)=1$

or $k^2=1[\because c^2+d^2=1]$

$\therefore k=1or-1$

When k = 1, a = d, b = -c and when k = -1, a = -d, b = c

Now $|\omega_1|=\sqrt{a^2+c^2}=\sqrt{a^2+b^2}=1$ $[\because c=-b]$

$|\omega_2|=\sqrt{b^2+d^2}=\sqrt{b^2+a^2}=1$ $[\because d=a]$

$\omega_1\bar{\omega}_2=(a+ic)(b-id)$ $\therefore$ Re $\omega_1\omega_2=ab+cd$

= - dc + cd = 0 $[\because a=d,b=-c]$ or a = -d , b = c

and Im $(\omega_1\bar{\omega}_2)$ = bc – ad = - (c^2+d^2) or $c^2+d^2=-1$ or 1

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13. $f\left(\frac{\pi}{2}\right)=0$

$f\left(\frac{\pi-}{2}\right)>0; f\left(\frac{\pi^+}{2}\right)>0 \Rightarrow$ f is not one – one

Similarly check in of y = 3 to include that f is not onto

14. $P=\left(at_1^2,2at_1\right), a\left(-at_1^2,0\right)$

$R\equiv\left(at_2^2,2at_2\right)$ where $t_2=-t_1-\frac{2}{t_1}$

15. Define $y=2^x$. Then we can define the quadratic as $y^2-2y+1-\frac{1}{k^2}$. Through quadratic formula or inspection, we notice that this quadratic can be factored as $\left(y-\left(1-\frac{1}{k}\right)\right)\left(y-\left(1+\frac{1}{k}\right)\right)$. Hence, $y=1\pm\frac{1}{k}$. Thus, $2^x=1\pm\frac{1}{k}\rightarrow x=\log_2\left(1\pm\frac{1}{k}\right)$.

Note that the sum of the two solutions to a single equation is

$\log_2\left(1-\frac{1}{k^2}\right)=\log_2\left(\frac{k^2-1}{k^2}\right)=\log_2\left(\frac{(k-1)(k+1)}{k^2}\right)$. The sum of all solutions to the equations is then

$N=\log_2\left(\frac{1.3}{2^2}\right)+\log_2\left(\frac{2.4}{3^2}\right)+\ldots+\log_2\left(\frac{2022.2024}{2023^2}\right)$

$=\log_2\left(\frac{1\cdot3}{2^2}\cdot\frac{2\cdot4}{3^2}\cdot\ldots\cdot\frac{2022\cdot2024}{2023^2}\right)$

$=\log_2\left(\frac{1\cdot2024}{2\cdot2023}\right)=\log_2\left(\frac{1012}{2023}\right)$ We have $=2^N=\frac{1012}{2023}$

17. $g(a)=4\int_{b^{1/4}}^{a^{1/4}} f(t^4)t^3dt$

Let $t^4=z\Rightarrow 4t^3dt=dz$

$g(a)=\int_a^b f(z)dz \quad g(b)=\int_a^b f(t)dt$

$g'(c)=\frac{g(b)-g(a)}{(b-a)}=\frac{2\int_a^b f(t)dt}{(b-a)}$

$\mu=2$

18. Let P (-1, y_1), $A(t_1)$ and $B(t_2)$, then $t_1t_2=-1, t_1+t_2=y_1$

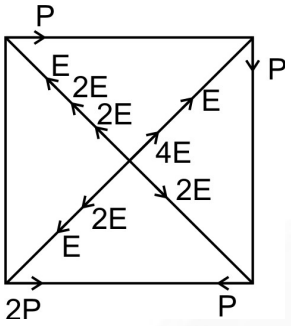
If circumcentre be (h, k), then h = $\frac{t_1^2+t_2^2}{2}$ and $k=y_1$

$\Rightarrow 2h=y_1^2+2\Rightarrow$ locus is $y^2=2x-2$

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PHYSICS

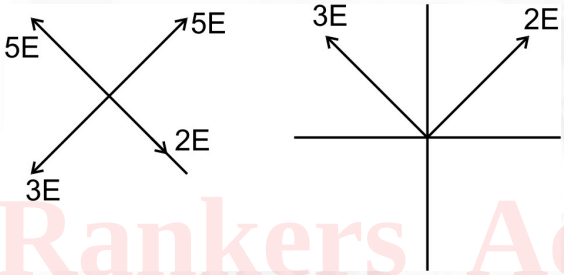
19.



$$E_r = \frac{2KP\cos 45^0}{r^3}$$

$$E_r = \frac{2KP}{\sqrt{2}r^3} = 2E \qquad E_\theta = \frac{KP\sin 45^0}{r^3}$$

$$E_\theta = \frac{KP}{\sqrt{2}r^3} = E$$



$$\overrightarrow{E_{net}} = \frac{-3E}{\sqrt{2}}\hat{i} + \frac{3E}{\sqrt{2}}\hat{j} + \frac{2E}{\sqrt{2}}\hat{i} + \frac{2E}{\sqrt{2}}\hat{j}$$

$$\vec{E} = -\frac{E}{\sqrt{2}}\hat{i} + \frac{5E}{\sqrt{2}}\hat{j} = -\frac{1}{\sqrt{2}}\frac{KP}{\sqrt{2}}\hat{i} + \frac{5}{\sqrt{2}}\frac{KP}{\sqrt{2}r^3}\hat{j}$$

$$= \frac{KP}{r^3}\left(-\frac{1}{2}\hat{i} + \frac{5}{2}\hat{j}\right) = \frac{KP}{r^3}\left(x\hat{i} + y\hat{j}\right) \qquad \therefore x + y = -\frac{1}{2} + \frac{5}{2} = \frac{4}{2} = 2$$

20. $$U_1 = \frac{1}{2}Li^2 = \frac{1}{2}.L.\left[\frac{\varepsilon}{R}\left(1 - e^{-1}\right)\right]^2$$

$$U_1 = \frac{1}{2}L.\frac{\varepsilon^2}{R^2}\left(1 - \frac{1}{e}\right)^2 \qquad \ldots\ldots\ldots\ldots\ldots\ldots (i)$$

$$U_2 = \int i^2 R dt = R.\frac{\varepsilon^2}{R^2}\int\left(1 - e^{-\frac{t}{\tau}}\right)^2 dt \qquad U_2 = \frac{\varepsilon^2 L}{R^2}\left[\frac{2}{e} - \frac{1}{2e^2} - \frac{1}{2}\right]$$

$$\frac{U_1}{U_2} = \frac{1}{2}\frac{\left(1 - \frac{1}{e}\right)^2}{\left(\frac{2}{e} - \frac{1}{2e^2} - \frac{1}{2}\right)}$$

21. Let ℓ_1 and ℓ_2 be the final length of the two parts, then from gas equation

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$$\frac{P_0A\ell_0}{T_0}=\frac{PA\ell_1}{T_1}=\frac{PA\ell_2}{T_2} \quad \ldots\ldots\ldots\ldots\ldots\ldots(i)$$

Considering the equilibrium of the piston in initial and final states, we get

$P_0A=kx_0, PA=kx$.

$$\Rightarrow \frac{P}{P_0}=\frac{x}{x_0} \quad \ldots\ldots\ldots\ldots \text{(ii)}$$

decrease in the length of spring = total increase in the length of the two chambers

$x-x0=\ell_1+\ell_2-2\ell_0 \quad \ldots\ldots\ldots\ldots$ (iii)

from relation (i)

$$\ell_1=\frac{P_0T_1\ell_0}{PT_0}, \ell_2=\frac{P_0T_2\ell_0}{PT_0}$$

using (ii)

$$\ell_1=\frac{x_0T_1\ell_0}{xT_0}, \ell_2=\frac{x_0T_2\ell_0}{xT_0}$$

putting these in (iii)

$$x-x_0=\frac{x_0\ell_0}{xT_0}(T_1+T_2)-2\ell_0$$

putting values and solving, we get $x=\frac{3}{20}m$

22. compression in rod $A=\frac{mg}{3}$; young modulus $=Y_A$

compression in another rod $=\frac{2mg}{3}$; young modulus $=Y_B$

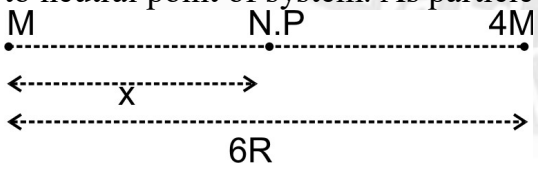
Since compression in both rod is same

$$\frac{\frac{mg}{3S}}{Y_A}=\frac{\frac{2mg}{3S}}{Y_B}\Rightarrow\frac{Y_B}{Y_A}\Rightarrow 2$$

Given

$$Y_A\alpha_A\Delta T=3Y_B\alpha_B\Delta T \qquad \frac{\alpha_A}{\alpha_B}=\frac{3Y_B}{Y_A}=6$$

23. Consider mass of the particle is m. We have to give speed to particle so that it can reach to neutral point of system. As particle crosses NP it will be pulled by 4M.

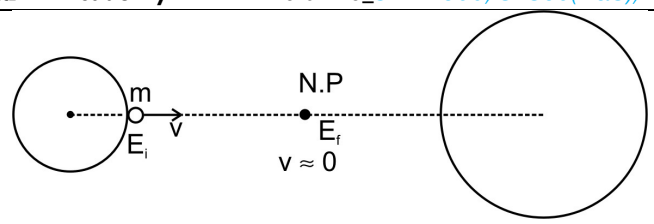


At N.P, gravitational field is zero.

Therefore, $\frac{GM}{x^2}=\frac{G\cdot 4M}{(6R-x)^2}\Rightarrow\frac{1}{x}=\frac{2}{6R-x}\Rightarrow x=2R$

Now applying energy conservation for initial and final positions.

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$X = 3$

24. Path difference produced by glass slab $= \left(1-\frac{\mu_2}{\mu_1}\right)t$

$\mu_2 \rightarrow$ R.I. of film

$\mu_1 \rightarrow$ R.I. of medium

Now, $\Delta x = \left(1-\frac{\mu_2}{\mu_1}\right).t = \frac{n.\lambda}{\mu}$

$\Rightarrow \left(1-\frac{2}{3/2}\right)t = \frac{n.\lambda}{\mu} \qquad \Rightarrow \left(1-\frac{2}{3}\right)t = \frac{n.\lambda.2}{3} \Rightarrow \frac{t}{3} = \frac{2n\lambda}{3}$

$\Rightarrow n = \frac{t}{2\lambda} = \frac{4\times10^{-6}}{2\times4000\times10^{-7}} \qquad = \frac{4\times10}{2\times4\times} = \frac{40}{8} = 5$

25. At half power frequencies,

$Z = R\sqrt{2} \Rightarrow |X_L - X_C| = R$

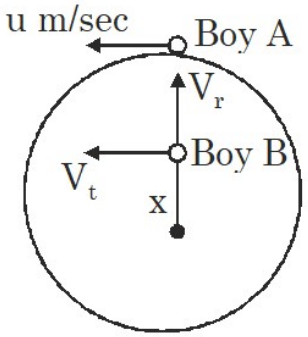
$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2 \Rightarrow \omega^2L^2 + \frac{1}{\omega^2C^2} - \frac{2L}{C} = R^2$

$\omega^2L^2C^2 - \omega^2\left(\frac{2L}{C} + R^2\right)C^2 + 1 = 0$

The product of roots is

$\omega_1^2\omega_2^2 = \frac{1}{L^2C^2} = \omega_1\omega_2 = \frac{1}{LC} = \omega_0^2 \Rightarrow \omega_0 = \sqrt{\omega_1\omega_2} = 60\text{rad/s} \Rightarrow \omega_0/10 = 6$

26. at any time t $\omega_A = \omega_B = \omega / R$



$v_t = \omega Bx$

$v_t = \frac{u}{R}x$

$v_b = u$

$\sqrt{V_t^2 + V_r^2} = u \qquad V_r = \frac{dx}{dt}$

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$$\left(\frac{u}{R}\right)^2 x^2 + \left(\frac{dx}{dt}\right)^2 = u^2$$

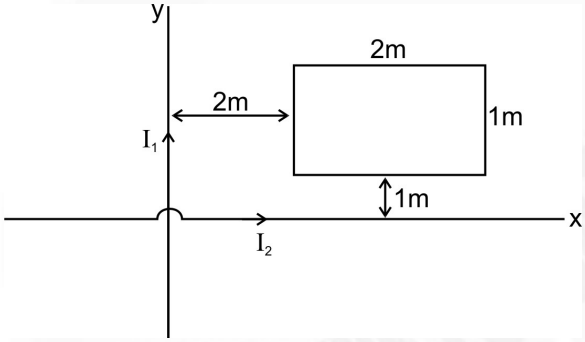
$$\left(\frac{dx}{dt}\right)^2 = u^2\left[1 - \frac{x^2}{R^2}\right]$$

$$\int_0^R \frac{dx}{\sqrt{1-\frac{x^2}{R^2}}} = u\int_0^t dt \qquad R\left[\sin^{-1}\frac{x}{R}\right]_0^R = ut$$

$$\frac{28\times\frac{22}{7}}{2} = ut \qquad t = 11\,\text{sec.}$$

$$10 + x = 11 \qquad x = 1$$

27.



$$I_2 = I_0\cos\left(\omega t + \frac{\pi}{4}\right) \qquad \phi_1 = \frac{\mu_0 I_1}{2\pi}\int_2^4 \frac{1dx}{x} = \frac{\mu_0 I_1}{2\pi}\ell n2$$

$$\phi_2 = \frac{\mu_0 I_2}{2\pi}\int_1^2 \frac{2dy}{y} = \frac{\mu_0 I_2}{\pi}\ell n2 \qquad \phi_{Total} = \frac{\mu_0 I_2 \ell n2}{2\pi} - \frac{\mu_0 I_2}{\pi}\ell n2$$

$$= \frac{\mu_0 \ell n2}{\pi} - \left[\frac{I_2}{2} - I_2\right] \qquad = \frac{\mu_0 \ell n2}{\pi}\left[I_0\sin\left(\omega t + \frac{\pi}{4}\right) - I_0\cos\left(\omega t + \frac{\pi}{4}\right)\right]$$

$$= \frac{\mu_0 \ell n2 I_0}{\pi}\left[\sin\left(\omega t + \frac{\pi}{4}\right) - \cos\left(\omega t + \frac{\pi}{4}\right)\right]$$

$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0 I_0 \omega \ell n2}{\pi}\left[\cos\left(\omega t + \frac{\pi}{4}\right) + \sin\left(\omega t + \frac{\pi}{4}\right)\right]$$

$$\varepsilon_{max} = \frac{\sqrt{2}\mu_0 I_0 \omega \ell n2}{\pi}$$

$$Q_{max} = \frac{\sqrt{2}\mu_0 I_0 \omega \ell n2}{\pi} C$$

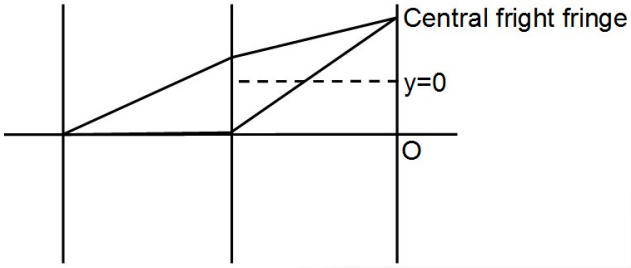
$$Q = CE$$

$$\frac{dQ}{dt} = C\varepsilon = C\frac{\mu_0 I_0 \omega^2 \ell n2}{\pi}\left[-\sin\left(\omega t + \frac{\pi}{4}\right) + \cos\left(\omega t + \frac{\pi}{4}\right)\right]$$

Current in frame at $t = \frac{\pi}{\omega}$ is zero.

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28.



Path difference $= SS_1 - SS_2 + S_1O - S_2O$

$$= \sqrt{D^2 + d^2} - D + \frac{dy}{3D} = D\left(1 + \frac{d^2}{2D^2}\right) - D + \frac{dy}{3D}$$

$$\therefore \Delta x = \frac{d^2}{2D} + \frac{dy}{3D} \text{ At O, Y} = \frac{d}{2}$$

$$\therefore \Delta x = \frac{d^2}{2D} + \frac{dy}{6D} = \lambda \therefore \Delta_{min} = \sqrt{\frac{3\lambda D}{2}}$$

For central maxima,

$\Delta x = 0 \therefore y = -\frac{3d}{2} \therefore$ Distance from O is $\frac{d}{2} + \frac{3d}{2}$

$= 2d_{min} = \sqrt{6\lambda D}$

29. $h_0 = \frac{h}{2}$

$$y = \frac{\omega^2 R^2}{2g} \Rightarrow \frac{h}{2} = \frac{\omega^2 R^2}{2g}$$

$$\omega = \sqrt{\frac{gh}{R^2}}$$

$$\frac{dp}{dx} = \rho\omega^2 x$$

$$\frac{dp}{dx} = -\rho g$$

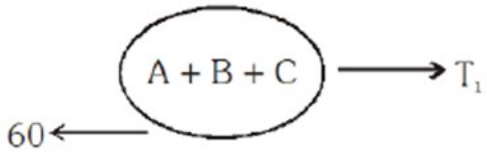
30. $fo_1 = 50$

$fo_2 = 100$

$fo_3 = 60$

There is a relative motion between C and D.

A + B + C] blocks move together.



$T_1 - 60 = 30a_1$ .....(1)

$100 - T_1 = 10a_1$ .....(2)

$a_1 = 1\,m/s^2$

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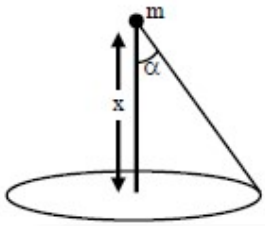
$E: 500 - T_2 = 50a_2$ .....(3)

$D: T_2 - 60 = 10a_2$ .....(4)

$a_2 = \frac{22}{3}$

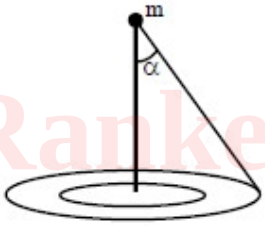
$T_2 = \frac{400}{3}N$

33.



$g = 2\pi G\sigma[1 - \cos\propto]$

$g = 2\pi G\sigma\left[1 - \frac{x}{\sqrt{R^2 + x^2}}\right] = 2\pi G\left[1 - \frac{x}{R}\right]$

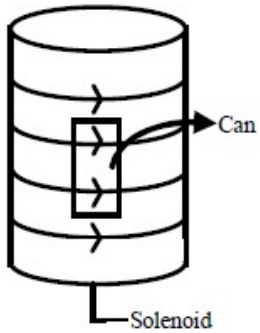


$g = 2\pi G\sigma\left[\left(1 - \frac{x}{R}\right) - \left(1 - \frac{x}{R/2}\right)\right]$

$g = \frac{2\pi G\sigma}{R}x$

$T = \sqrt{\frac{2\pi R}{G\sigma}}$

34.

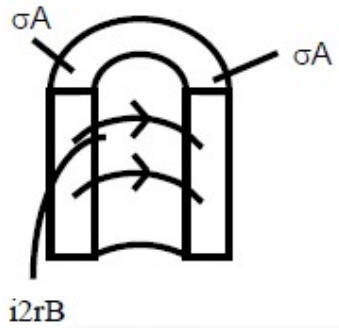


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Emf is included in the can

$$\text{Emf} = \frac{dB}{dt}\pi r^2 = K\pi r^2$$

$$R = \frac{\rho \times 2\pi r}{A}\, i = \frac{r}{2\rho}AK$$



From the diagram

$$i \times 2rB = 2\sigma A \; t = \frac{2\sigma}{r^2K^2}$$

35.



$$\vec{F}\text{ on platform} = m\vec{g} + mw^2\vec{r}$$

$$m\omega^2 r = \frac{\mu 5mg}{\sqrt{1+\mu^2}} \quad \omega_0 = 5 \text{ rad/s}$$

36. For perfect conductor R = 0 $\qquad E_{net} = 0$ in loop $\phi_{net} =$ constant

$$\phi_{net} = \phi_{self} + \phi_k + \phi_{ext}$$

$$= L_i + L_{ki} + \vec{B}.\vec{A} = L_{neti} + \vec{B}.\vec{A}$$

$$L_K = \frac{mI}{ne^2A}$$ (Where m mass of one electron, I = loop length)

As ϕ is constatn $\phi_i = \phi_f$

Initially $i = 0$ & $\vec{B} \perp \vec{A}$

$\phi_{net} = 0$

Finally $\phi = -BA' + L_{net}i$

$$i = \frac{B\pi r^2}{L_{net}} = \frac{B\pi r^2}{L + \frac{m \times 2\pi r}{ne^2A}}$$

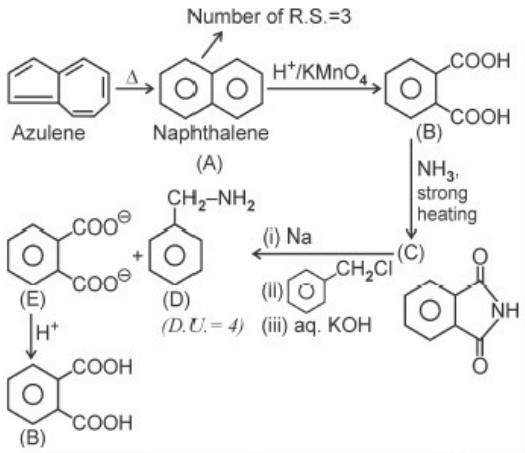
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CHEMISTRY

37. (I) $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$

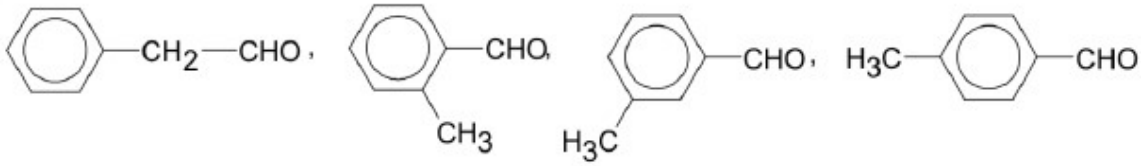
(II) $x_4 = 29$

38.



39. x = 7, y = 24, z = 7 x 18 = 126

40.



41. $2NaBH_4 + I_2 \rightarrow B_2H_6(g) + 2NaI + H_2(g)$

$\Rightarrow$ Total 6 moles of gaseous products

42. $Au^+ + 2CN^- \rightleftharpoons [Au(CN)_2]^-$

At time t a b c

$$\frac{[Au(CN)_2]^-}{[Au^+]+[Au(CN)_2]^-} = \frac{99}{100}$$

$$\Rightarrow \left[Au^+\right] = \frac{[Ag(CN)_2]^-}{99}$$

Given $4\times10^{28} = \frac{[Au(CN)_2]^-}{[Au^+][CN^-]^2}$

$$\Rightarrow [CN^-]^2 = \frac{[Au(CN)_2]^-}{[Au^+]\times 4\times10^{28}} = \frac{99}{4\times10^{28}}$$

$$\Rightarrow [CN^-] = 5\times10^{-14} = x\times10^{-y}$$

So $\frac{x+y}{y-x} = \frac{19}{9} = 2.11$

43. $1.\ Mn^{2+} + S_2O_8^{2-}$, $2.\ Mn^{2+} + NaBiO_3 + HNO_3$, $3.\ Mn^{2+} + PbO_2 + HNO_3$,

$4.\ K_2MnO_4 + Cl_2$

44. $K_2[NiCl_4]$ sp^3 paramagnetic

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$K_3[Fe(CN)_6]$	d^2sp^3 paramagnetic
$K_3[Cr(CN)_6]$	d^2sp^3 paramagnetic
$K_2[PtCl_4]$	dsp^2 diamagnetic
$Na_2[Fe(CN)_5NO]$	d^2sp^3 diamagnetic
$[Fe(H_2O)NO]SO_4$,	sp^3d^2 paramagnetic
$K_3[Co(C_2O_4)_3]$	d^2sp^3 diamagnetic
$K_4[Co(CN)_6]$	d^2sp^3 paramagnetic
$[Cu(NH_3)_4]SO_4$	dsp^2 paramagnetic

45. $Cu_2SO_4(aq) \rightarrow Cu^{+1}(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$ as $Cu^{+1}(aq)$ disproportionates in aqueous medium

$CO^{3+} \rightarrow CO^{2+}$ $E^o = +ve$

Cr^{2+} is a good reducing agent and evolve hydrogen gas from dilute acids.

46. $\frac{[B]}{[C]} = \frac{3K_1}{8K_1} = \frac{3}{8} = a$

$\frac{[C]}{[B]} = \frac{8K_1}{7.5K_1} = \frac{8}{7.5} = \frac{16}{15} = \beta$

47. The correct answer option is (B), (D)

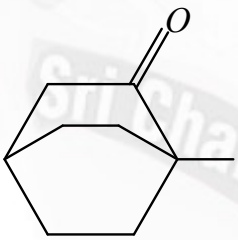
48. $A \rightleftharpoons P$ $\quad T_2 > T_1, \frac{\ell nk_1}{\ell nk_2} > \frac{T_2}{T_1}$

$T_1 \ell_n k_1 > T_2 \ell nk_2 \quad -\Delta G_1^o > \Delta G_2^o$

$\left(-\Delta H^o + T_1\Delta S^o\right) > \left(-\Delta H^o + T_2\Delta S^o\right)$

$T_1\Delta S^o > T_2\Delta S^o \quad \Rightarrow \Delta S^o < 0$

49. A.



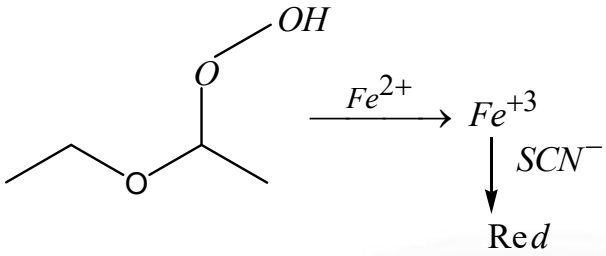
Phenol $\rightarrow$ *Phenol pthalein*

$\downarrow$ *base*

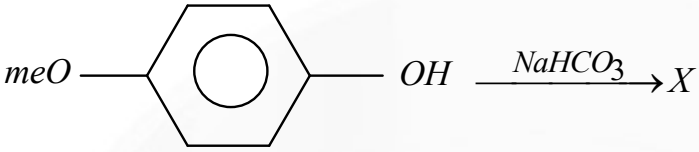
Pink color

B.

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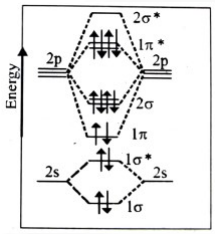
C.



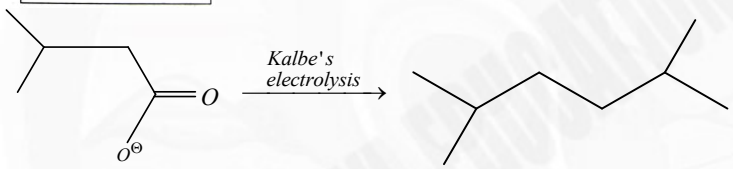
D.

51. Atomic number = 103 = Unniltrium

52.

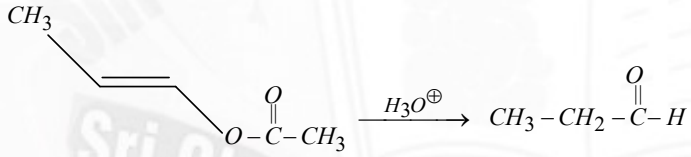


53.



54.





Fehling solution is reduced with aldehyde.

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